

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

VECTOR CALCULUS I: Vector fields and Line integral of functions

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0.1 Vector fields

Let $n \geq 2$ be an integer. Let $D \subset \mathbb{R}^n$. A vector function of the type $\vec{F} : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called an **n -dimensional vector field**. Therefore, the vector field $\vec{F}$ assigns to every point $(x_1, x_2, \dots, x_n) \in D$ a vector with n components

$$\vec{F}(x_1, x_2, \dots, x_n) = (F_1(x_1, x_2, \dots, x_n), F_2(x_1, x_2, \dots, x_n), \dots, F_n(x_1, x_2, \dots, x_n)).$$

How do we picture a vector field in $\mathbb{R}^2$ or $\mathbb{R}^3$? The best way to picture a vector field (in $\mathbb{R}^2$ for instance) is to draw the arrow representing the vector $\vec{F}(x, y) = (P(x, y), Q(x, y))$ starting at the point (x, y) .

Examples 0.1 Draw the vector field:

- (a) $\vec{F}(x, y) := (-y, x) = -y\vec{i} + x\vec{j}$
- (b) $\vec{F}(x, y) := -(x, y) = -x\vec{i} - y\vec{j}$
- (c) $\vec{F}(x, y, z) := x\vec{k}$.

Examples 0.2

- (a) **The gravitational force:** is the force generated by a particle of mass M , located at the origin and acting on another particle of mass m located at (x, y, z) . It is known according to **Newton's Law of gravitation** that such a vector field is defined as

$$\vec{F} : \mathbb{R}^3 \setminus \{\vec{0}\} \rightarrow \mathbb{R}^3, \quad \vec{F}(x, y, z) := -GmM \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$$

where G is called **gravitational constant**.

- (b) **The electric force:** is the force exerted by a charge Q located at the origin, on another charge q located at (x, y, z) . It is known that according to the so-called **Coulomb's Law** that such a vector field is defined as

$$\vec{F} : \mathbb{R}^3 \setminus \{\vec{0}\} \rightarrow \mathbb{R}^3, \quad \vec{F}(x, y, z) := \varepsilon_0 qQ \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$$

where ε_0 is called **electric constant**.

- (c) **Gradient vector fields:** If $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function which both partial derivative $f_x(x, y)$ and $f_y(x, y)$ at the every $(x, y) \in D$, then we recall that its gradient is the 2D vector field

$$\nabla f(x, y) = (f_x(x, y), f_y(x, y))$$

likewise if $f : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ then

$$\nabla f(x, y, z) := (f_x(x, y, z), f_y(x, y, z), f_z(x, y, z)).$$

∇f is called a gradient vector field.

We say that a vector field $F : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a **gradient vector field** if there exists $f : D \rightarrow \mathbb{R}$ differentiable such that

$$\vec{F}(x_1, x_2, \dots, x_n) = \nabla f(x_1, x_2, \dots, x_n) \quad \forall (x_1, x_2, \dots, x_n) \in D.$$

Such a vector field is also called a **conservative vector field** and a scalar functions f such that $\vec{F} = \nabla f$ is called a **potential** of the vector field $\vec{F}$. Notice that, for a conservative vector field $\vec{F}$, there corresponds infinitely potentials. The following terminology is often used in Physics: **the vector field $\vec{F}$ is derived from a scalar potential f .**

Exercise 0.3 Show that the vector field

$$\vec{F}(x, y) := \left(-\frac{2xy}{(1+x^2)^2}, \frac{1}{1+x^2} \right)$$

is conservative.

Solution: We seek a differentiable function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $\vec{F} = \nabla f$. That is,

$$\begin{cases} f_x(x, y) = -\frac{2xy}{(1+x^2)^2}, \\ f_y(x, y) = \frac{1}{1+x^2} \end{cases} \quad (1)$$

We get from the second equation that $f(x, y) = \frac{y}{1+x^2} + g(x)$. Now, we differentiate this w.r.t. x and get

$$f_x(x, y) = -\frac{2xy}{(1+x^2)^2} + g'(x)$$

which using the first equation in (1) gives

$$g'(x) = 0 \quad \Rightarrow \quad g(x) = C \quad (\text{constant}).$$

Hence, we obtain that

$$f(x, y) = \frac{y}{1+x^2} + C \quad \forall (x, y) \in \mathbb{R}^2$$

0.2 Line integrals

0.2.1 Line integrals for scalar-valued functions

In this section, we present a notion of a single integral over a parametric curve. Before we give the definition, let us consider the problem of finding the mass of a beam represented here by a curve $\mathcal{C}$ described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^3$. Assume that the beam has mass density per unit length given by the continuous function $\rho: \mathcal{C} \rightarrow \mathbb{R}$. Then the mass m_ε of a little piece of the beam with length $\varepsilon > 0$ and containing the point (x_0, y_0, z_0) is $\approx \rho(x_0, y_0, z_0)\varepsilon$. Now in order to find the mass of the beam, we can proceed by approximation as follows: we consider a partition $P := \{a = t_0 < t_1 < t_2, \dots < t_n = b\}$ to which there corresponds a set of points $\vec{r}(t_0), \vec{r}(t_1), \dots, \vec{r}(t_n)$ on the curve on the curve $\mathcal{C}$. We get that

$$\begin{aligned} m(\mathcal{C}) &= \sum_{i=1}^n m(C_i) \approx \sum_{i=1}^n \rho(x_i, y_i, z_i) \Delta s_i \\ &\approx \sum_{i=1}^n \rho(x_i, y_i, z_i) \|\vec{r}(t_i) - \vec{r}(t_{i-1})\| \approx \sum_{i=1}^n \rho(\vec{r}(t_i)) \|\vec{r}'(t_i)\| \Delta t_i \end{aligned}$$

Now by refinining the partition, since the last sum is the Riemann sum of the function $t \rightarrow \rho(\vec{r}(t))\|\vec{r}'(t)\|$, we obtain that

$$M(\mathcal{C}) = \int_a^b \rho(\vec{r}(t)) \|\vec{r}'(t)\| dt.$$

In general, the definition of the line integral an n -dimensional vector is given below.

Definition 0.4 (The linear integral of a scalar-valued function)

Let $f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ a continuous function. Let $\mathcal{C}$ be a regular curve described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$ which takes its values in the set D . Then, the linear integral of the function f along the curve $\mathcal{C}$ is defined as

$$\int_{\mathcal{C}} f(x, y) ds = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| dt. \quad (2)$$

If $\mathcal{C}$ is a piecewise regular curve, that is, $\mathcal{C}$ is a finite union of regular curves $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_n$ with $\mathcal{C}_i \cap \mathcal{C}_{i+1} = \{\text{single point}\}$, then

$$\int_{\mathcal{C}} f(x, y) ds = \sum_{i=1}^n \int_{\mathcal{C}_i} f(x, y) ds \quad (3)$$

Example 0.5 Evaluate the line integral $\int_{\mathcal{C}} 2x^2 ds$ where $\mathcal{C}$ is the arc of the circle centered at the origin with radius 2, from the point $(\sqrt{2}, -\sqrt{2})$ to $(-1, -\sqrt{3})$. The curve is described the vector function

$$\vec{r}(t) := (2 \cos t, 2 \sin t), \quad -\frac{\pi}{4} \leq t \leq \frac{4\pi}{3}.$$

So,

$$\vec{r}'(t) = (-2 \sin t, 2 \cos t) \Rightarrow \|\vec{r}'(t)\| = 2.$$

Hence

$$\begin{aligned}\int_{\mathcal{C}} 2x^2 ds &= \int_{-\frac{\pi}{4}}^{\frac{4\pi}{3}} 16 \cos^2 t \, dt = \int_{-\frac{\pi}{4}}^{\frac{4\pi}{3}} 8(\cos(2t) + 1) \, dt = \left[4 \sin(2t) + 8t \right]_{-\frac{\pi}{4}}^{\frac{4\pi}{3}} \\ &= 4 \sin\left(\frac{8\pi}{3}\right) + \frac{32\pi}{3} + 4 + 2\pi = 4 \sin\left(\frac{2\pi}{3}\right) + \frac{32\pi}{3} + 4 + 2\pi = \frac{32\pi}{3} + 2\sqrt{3} + 4 + 2\pi.\end{aligned}$$

Remark 0.6 Geometrically the line integral $\int_{\mathcal{C}} f ds$ can be interpreted as the area of a piece of some cylindrical surface. For instance, consider the curve $\mathcal{C}$ described by the vector function $\vec{r}: [\pi, 2\pi] \rightarrow \mathbb{R}^2$ defined by $\vec{r}(t) := (\cos t, \sin t)$ and consider the function $f(x, y) := -y$. Sketch the area corresponding to $\int_{\mathcal{C}} f ds$.

Exercises 0.7

1. Consider the curve $\mathcal{C}$ described by the vector function $\vec{r}: [1, e] \rightarrow \mathbb{R}^2$ defined by

$$\vec{r}(t) := (t + \ln t, t - \ln t), \quad 1 \leq t \leq e.$$

- (a) Show that the curve $\mathcal{C}$ is regular.
- (b) Find the parametric equations of the tangent lines to $\mathcal{C}$ at the points $(1, 1)$ and $(e + 1, e - 1)$.

- (c) Evaluate the line integral $\int_{\mathcal{C}} (x + y)^2 ds$. **Answer:** $\boxed{\frac{4\sqrt{2}}{3} \left[(1 + e^2)^{3/2} - 2\sqrt{2} \right]}$.

2. Let $\mathcal{C}$ be the ellipse obtained by intersecting the cylinder $x^2 + y^2 = 1$ with the plane $z = x$.

- (a) Write down a parametric representation of the ellipse $\mathcal{C}$.

- (b) Evaluate the line integral $\int_{\mathcal{C}} \sqrt{1 + y^2} \, ds$. **Answer:** $\boxed{3\pi}$.

3. Let $\mathcal{C}$ be the ellipse obtained by intersecting the cylinder $x^2 + y^2 = 1$ with the plane $x + y = z$.

- (a) Write down a parametric representation of the ellipse $\mathcal{C}$.

- (b) Evaluate the line integral $\int_{\mathcal{C}} \sqrt{3(x^2 + y^2) - z^2} \, ds$. **Answer:** $\boxed{4\pi}$.

**The next set of notes will be on
the line integrals of vector fields!**